Produce projects plans & Present project recommendations, planning a Computing Project for Cyber Security

Thet Htoo Zin (Pearson ID: UE22116)
GUSTO COLLEGE

Table of Contents

Introduction 3
Primary Research: Survey Data 3
Demographic Summary 3
IoT Usage & Landscape 5
Authentication Habits Password Behaviors 7
Key Behavioral Correlations 9
Awareness & Threat Perception 9
Why Users Keep Default or Old Passwords 11
Worst-Case Condition That Users are Concerned 11
Localized Challenges (Myanmar Context) 13
Real-World Incidents from Myanmar 14
Secondary Research 14
Authentication Vulnerabilities in IoT 14
Empirical Evidence to Determine Platform Vulnerabilities 15
Features and Operational Areas: IoT Management Platform Sector 16
The Role of Stakeholders and Their Impact 16
Internal stakeholders 16
External stakeholders 17
Analyze Data and Information from Primary and Secondary sources 17
Primary Research 17
Secondary Source Analysis 18
Key Challenges to Business Success 18
Project Planning: Work Breakdown Structure and Gantt Chart 19
Work Breakdown Structure 19
Gantt Chart 21
Risks and Benefits of the Proposed Project 22
Project Benefits 22
Project Risks 22
Assessment of Project Recommendations Against Organizational Needs 23

Evaluation of Project Planning Recommendations and Research Reliability 25
Assessment of Project Planning Alignment with Organizational Needs .. **Error! Bookmark not defined.**
Evaluation of Research Accuracy and Research Reliability 26
Overall Evaluation 27
Conclusion 28
References 28

Introduction

IoT devices make our lives easier. They also bring security risks. A major problem is weak authentication. Authentication is the key to protect unauthorized access. Most people use default password (common password). That make vulnerable to cyber-attacks.

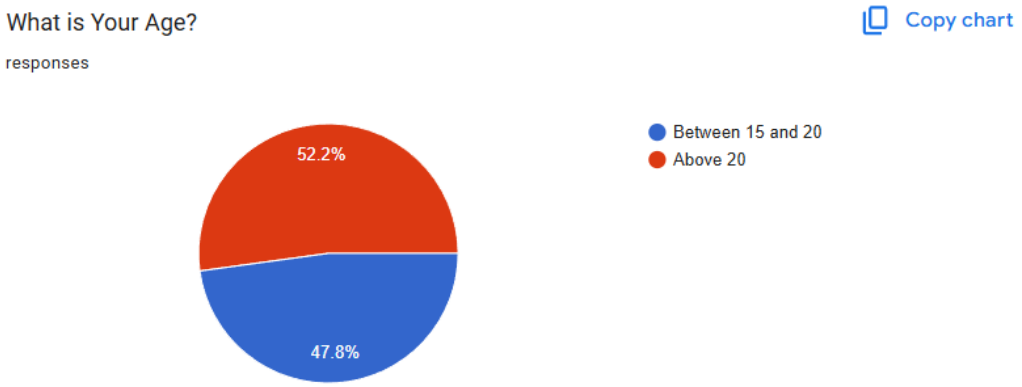
Although many people have security awareness, usability problem, credential overload and budget can limit their use. In Myanmar, there are challenges including lacks of electricity, low internet quality. This project explores issues to find ways to improve IoT devices security.

Primary Research: Survey Data

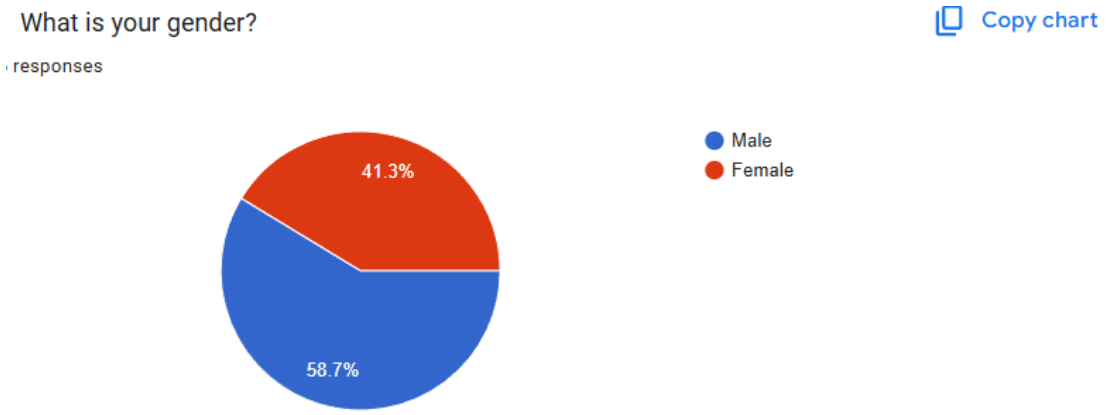
The primary research was carried out through a mixed methods survey approach, which involved quantitative and qualitative methods to gather in-depth insights into the use of IoT authentication. This method allowed the research to reflect both the observable behavior and the user experiences in the context.

Demographic Analysis

Age distribution: 52.2% (24 respondents) are above 20, and 47.8% (22 respondents) are between 15 and 20.



Gender: 58.7% male (27 respondents) and 41.3% female (19 respondents).



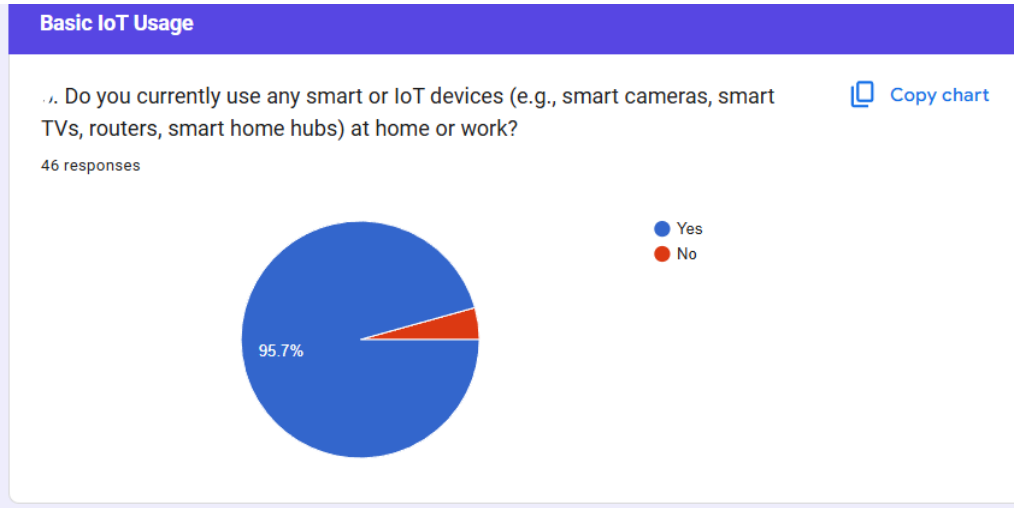
Respondents Status: Students (73.9% / 34 respondents) and employed individuals (21.7% / 10 respondents), and self-employed individuals (4.3% / 2 respondents).



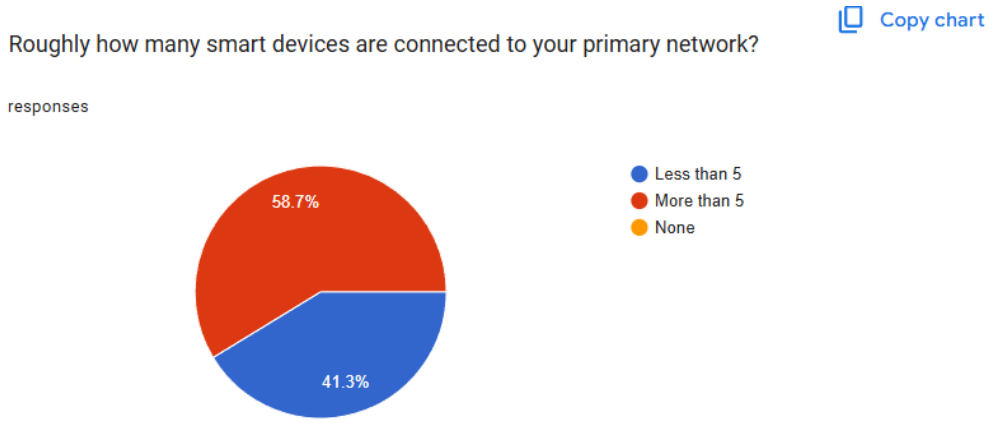
Quantitative Analysis

IoT Usage & Landscape

Adoption Rate: (95.7%) use IoT devices at home or work.

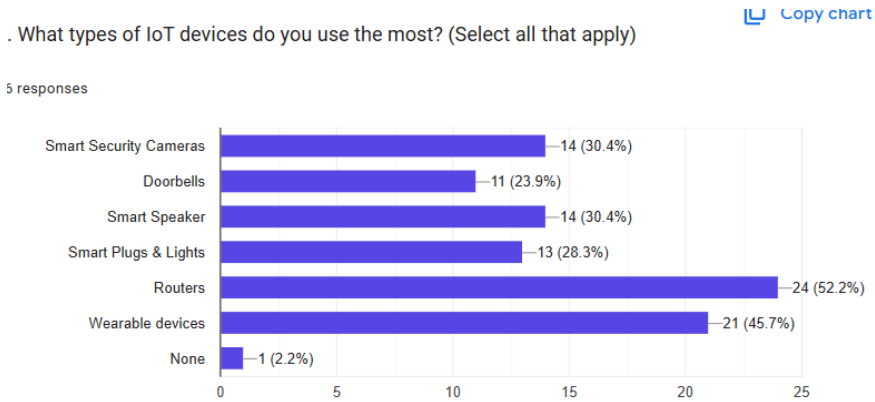


Network Density: 58.7% use more than 5 devices connected to their main network and 41.3% use less than 5 devices.



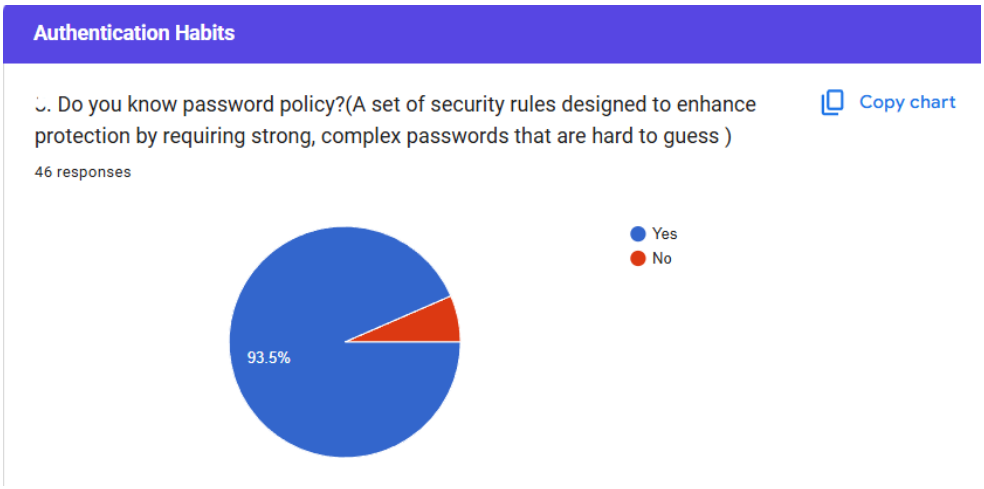
Most Popular Devices: Routers, Wearable devices and Smart speakers/security cameras:

1. Routers (24 selections)
2. Wearable Devices (21 selections)
3. Smart Security Cameras (14 selections)
4. Smart Speakers (14 selections)
5. Smart Plugs & Lights (13 selections)
6. Doorbells (11 selections)

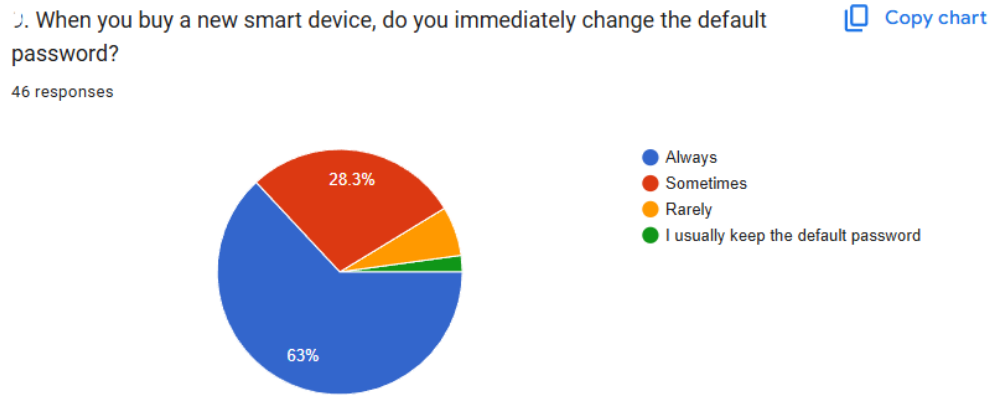


Authentication Habits Password Behaviors

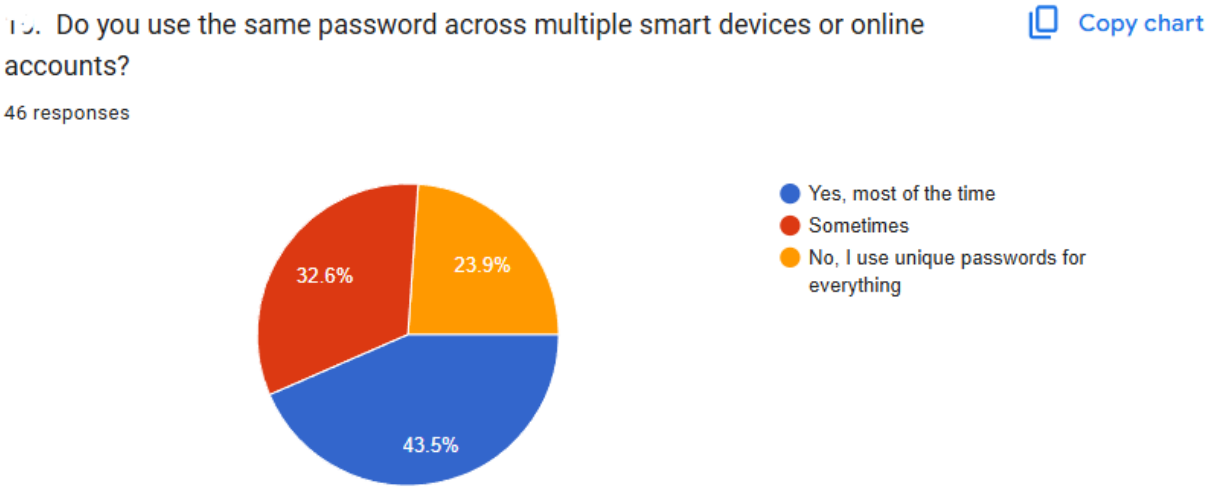
Policy Awareness: 93.5% respond that they understand password policy.



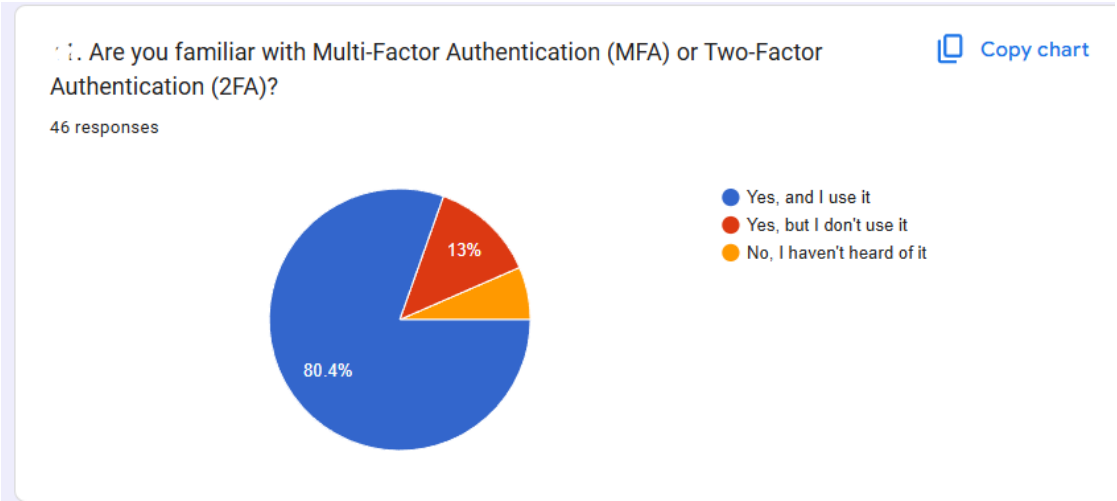
Default Passwords: 63.0% always change default passwords on purchase. 28.3% sometimes change default passwords on purchase. 8.7% rarely or never change default passwords on purchase.



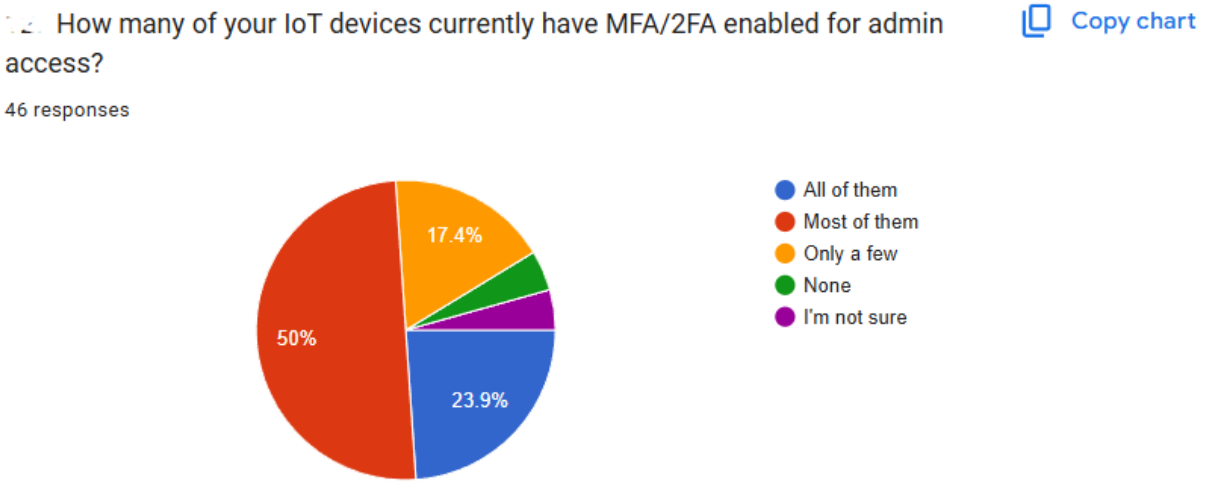
Password Reuse: Even default passwords are changed, password reuse is common. 43.5% reuses the same credentials most of the time across different accounts/devices and 32.6% reuse them sometimes. Just 23.9% use unique credentials all the time.



Multi-Factor Authentication awareness: 80.4% (37 respondents) of users are aware of and use MFA/2FA.



Most of their IoT devices are configured with MFA for administrative control.



Key Behavioral Correlations

70.0% of employed people always change default passwords. 61.8% of students do. 32.4% students do it Sometimes.

61.5% use MFA. On the other hand, fewer than 5 users show more polarization. 27.8% have managed to protect all devices.

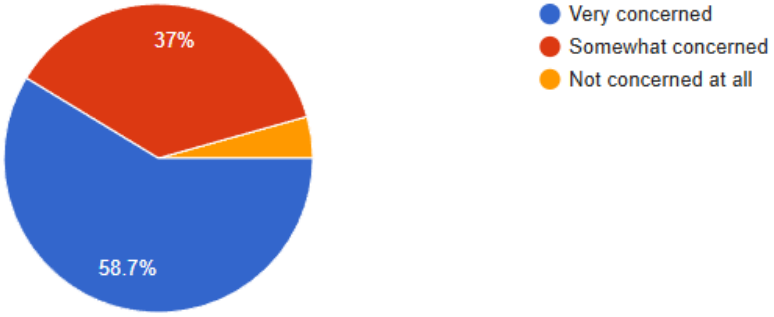
Awareness & Threat Perception

58.7% are very concerned and 37.0% are somewhat concerned.

How concerned are you about unauthorized users accessing your smart device data (e.g., viewing live camera feeds or changing device settings)?

Copy chart

46 responses

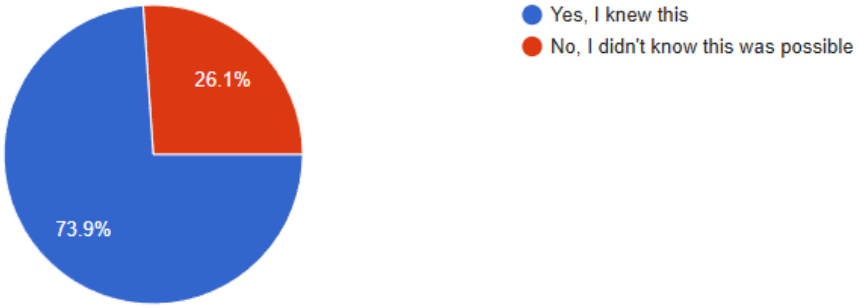


Botnet Awareness: 73.9% know that IoT systems can be turned into a botnet.

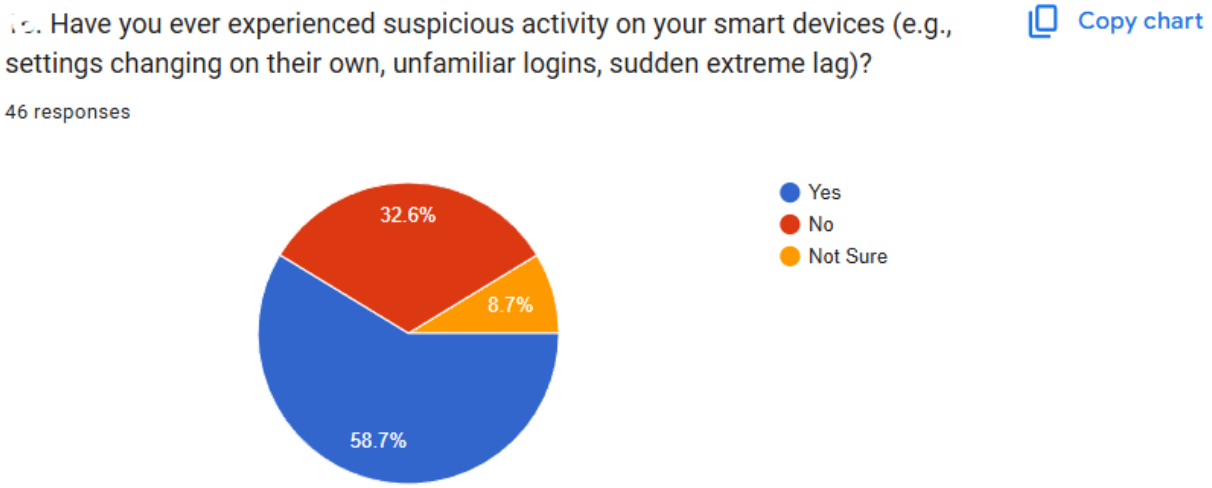
Were you aware that hackers can hijack poorly secured home devices and force them into a "botnet" to launch massive cyber attacks against other websites?

Copy chart

46 responses



Suspicious Encounters: 58.7% have experienced suspicious activity on their IoT devices.



Qualitative Analysis

Why Users Keep Default or Old Passwords

Cognitive Fatigue / Memory Overhead: Users say they fear that they'll forget the unique passwords.

Setup Configurations: Users say that the native setup process in applications is sometimes too complicated. Users sometimes skip security updates to stop their devices from system errors.

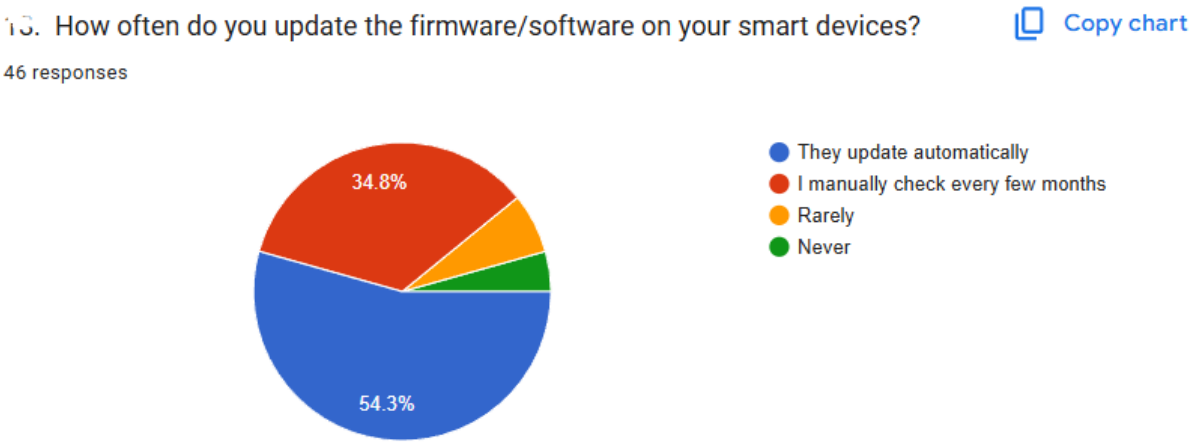
Worst-Case Condition That Users are Concerned

Surveillance & Blackmail: The most common fear is an attacker hijacking smart security cameras to capture private, home moments and use the records footage for blackmail.

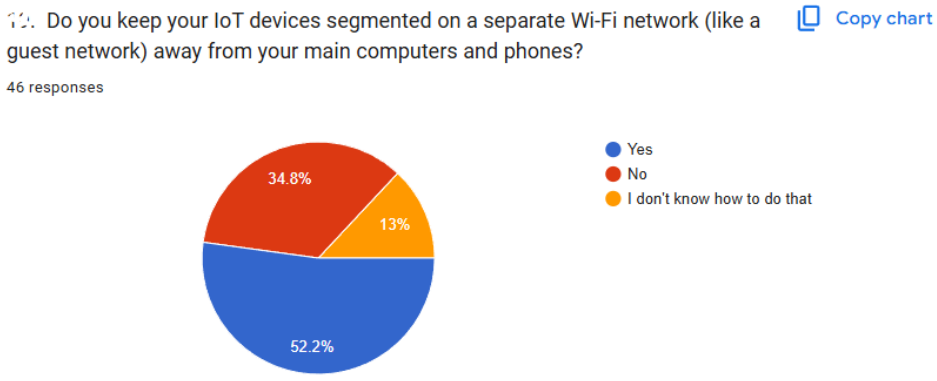
Gateway Pivoting: Users are concerned that if they are able to penetrate a small device (such as router), it will be a gateway to go around their home or office network, getting access to personal computers and stealing banking information or documents.

Disagreements over the appropriateness of devices

Users are strict with vendors having unclear firmware management policies. Updates are vague, font have auto installation and contain technical menus that don't explain the real-world impact of toggling different selections.

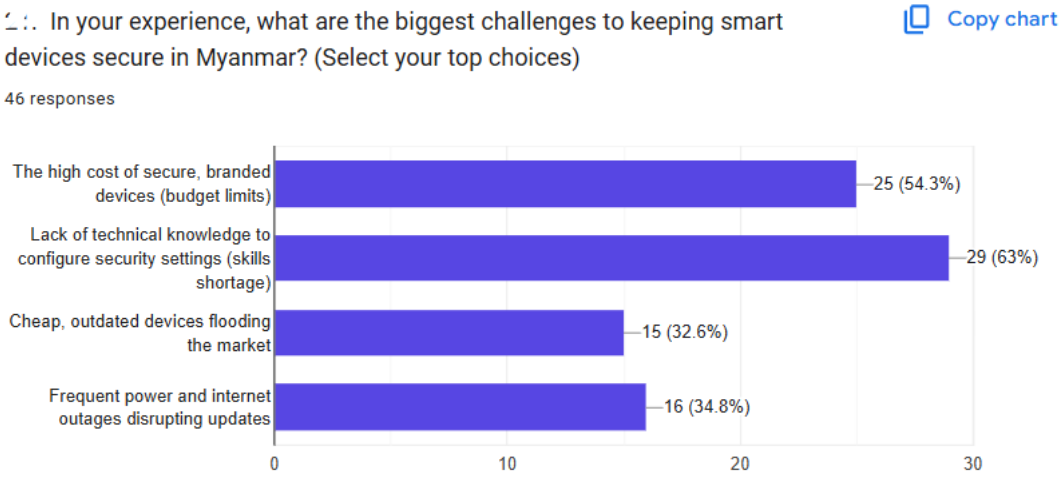


Guest Wi-Fi Network: Creating a separate Wi-Fi network for IoT devices to keep them separate from the main network is emphasized as being very confusing and awkward to set up for non-technical users.

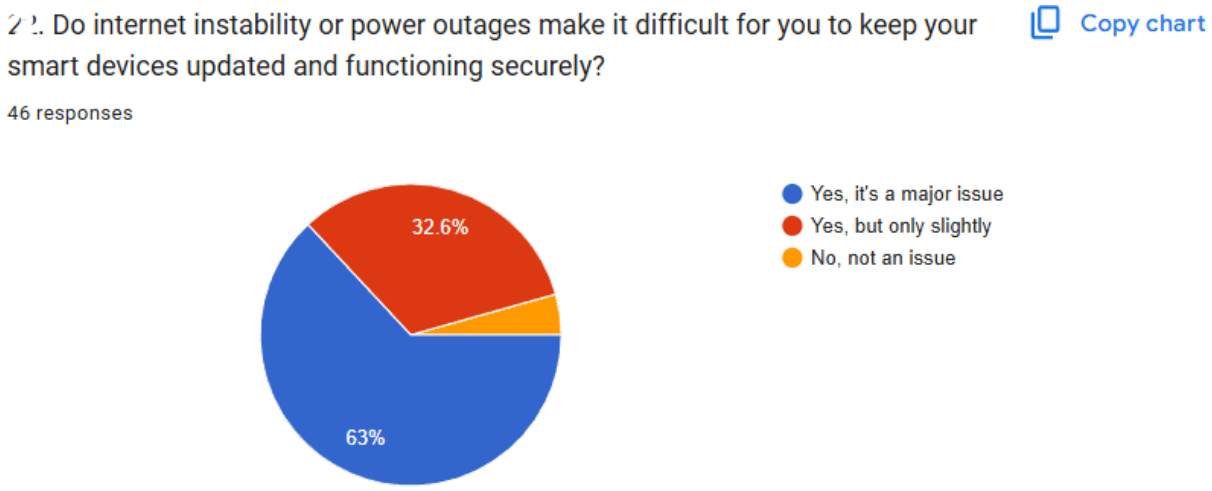


Challenges in Myanmar

The biggest issue highlights lacks of technical skills and knowledge to configure security settings. The high number of budget constraints allows for the implementation of not-branded, lower quality or even obsolete equipment, which is readily available in the local market.



Infrastructure Failure: Infrastructure stability is a security threat in the present time. 63.0% believe power outages and internet outages are a major issue on maintaining secure devices.



Real-World Incidents from Myanmar

Here are some case studies that illustrate the effects of local infrastructure on cybersecurity:

The Bricked Firmware Vulnerability: One of the respondents explained trying to perform a normal update of the home router's firmware, but the power went out during the process. After power was restored, due to disconnections, it resulted in a full factory reset and left the device in an unpatched, insecure and extremely vulnerable state.

The Security Camera Failure: The second failure that was reported was due to a broken security camera software patch that was pushed during a rolling black-out. Otherwise, it will permanently brick the device and the importation cost and tech inflation in Myanmar is high then how to replace it with security.

Data Cost Penalty: Some respondents said that having their fixed broadband cut meant they need to use a cellular data plan to receive security updates. But the cost of local mobile data, however, is too high for the download of large firmware files, so that users purposely withhold essential security updates.

Autonomous Setting Resets: One user reported that their home router was constantly power cycling and resetting to the factory settings with no warning, even though the security configuration has been set.

Secondary Research

Authentication Vulnerabilities in IoT

According to their latest paper, lots of systematic literature reviews have been conducted, and they have found that one of the major weaknesses in the IoT infrastructure and therefore device security and data integrity is authentication (Ashwag Alotaibi , Huda Aldawghan and Ahmed Aljughaiman, 7 March 2025).. The basic authentications like username & password or Public Key Infrastructure (PKI)

can be used by IoT devices but it is difficult to implement/ use for resource constrained devices due to computational power and scalability issues. The different attack vectors against IoT authentication methods are described—Man-in-the-Middle attack, Replay attack, Credential theft attack and Denial-of-Service attack. The identified weaknesses highlight the need for authentication protocols that are lightweight and suitable for specific contexts, tailored for IoT systems in the context of smart cities.

Empirical Evidence to Determine Platform Vulnerabilities

In empirical testing of typical and authentic use of IoT management platforms, many types and categories of vulnerabilities in these platforms have been shown. Five of these 52 IoT management platforms (Bhaskar Tejaswi, Student Member, IEEE, Mohammad Mannan, and Amr Youssef, 26 Jul 2023) contained partial information secured by broken authentication vulnerability, 13 platforms suffered from complete information stolen through account takeover vulnerabilities and 2 platforms had remote code execution vulnerabilities. The vulnerabilities can be used by the attacker to perform critical operations such as activation of a SIM card, data forging, data transaction to the device and from it and arbitrary command running within the device. The results indicate the critical nature of the security assessment techniques, and the widely used authentication methods in the exchange of authentication methods in the IoT. This study emphasizes the importance of applying severe security assessment techniques and the level of authentication in all IoT platforms, irrespective of connected elements and the possible vulnerabilities of platform-wide attacks on users and connected elements.

Features and Operational Areas: IoT Management Platform Sector

The Internet of Things management platform sector includes companies that provide services for infrastructure of managing large scale deployments of connected devices in consumer, enterprise, and industrial applications. There are two predominant models. They are web hosted solutions managed by third parties and locally deployed solutions run by enterprises and managed themselves.

Real-time monitoring & diagnostics of operations, automated alerts on user-defined conditions, multi-user access control, and extensive reporting for usage tracking and billing provide support for these operational aspects. Users of these platforms range from small businesses that have a fleet of these devices to businesses with millions of connected devices such as multi-national companies. From smart cities to healthcare systems, industrial applications, transportation networks, home automation, the sector is wide ranging and the commercial potential of IoT management platform services is tremendous.

The Role of Stakeholders and Their Impact

Engaging stakeholders is essential to a successful research project and can impact every aspect of the project from execution to outcome and impact. Stakeholder engagement plays an important role in the success of the project from in and out of the project and effects project execution, project outcomes and project impact.

Internal stakeholders

The project manager and various internal stakeholders are responsible for providing guidance and driving the project for a specified timeline from various steps of the project to enable a systematic process of deliverables culminating to project completion. The technical knowledge and manual labor, which are carried out by research assistants or data analysts, are vital for collecting data, testing platforms and empirical analysis. The faculty advisor specializing in cybersecurity and IoT has a wealth of knowledge on research methodology, theory and academic standards

and the University admins can provide all sorts of infrastructure and secure testing platforms and sandboxes that would otherwise be impossible to have without their help. To engaging in academic dissemination, department coordinators help in paper submissions and promoting researches as well to make the project visible and influential.

External stakeholders

However, the legitimacy and applicability of the project is also heavily affected by external stakeholders. The academic peer reviewers and academic journals editors make the decision whether or not findings of the project will be published in legitimate journals, or whether or not findings of the project will be published in legitimate journals directly affecting the academic credibility and reach of the project. Industry security experts validate the methodology used and provide expert commentary on the significance of the research, along with providing expert commentary on potential impact of published research findings to manufacturers of IoT devices, who could be potential beneficiaries of vulnerability reports and potential stakeholders that may be affected by research results. In most cases, the research is funded and there is a sustainability point that is set as a performance goal and that could affect project priorities. Effective coordination across these diverse stakeholders and keeping in mind academic strictness, and ethical disclosure of practices, technical feasibility and relevance.

Analyze Data and Information from Primary and Secondary sources

Primary Research

The survey revealed that there is a huge awareness-action gap in the area of IoT security: 46 participants were asked. 73.9% of respondents implement Multi-Factor Authentication (MFA) and 93.5% have strong password policies that they have in place.73.9% of respondents implement Multi-Factor Authentication (MFA)

and 93.5% have strong password policies that are in place. While security is a concern, users have been seen to be more concerned about convenience than security, with one reason given for not changing default credentials being that they remembered their password. Moreover, 63% of respondents are affected by power instability in Myanmar, directly influencing the firmware updates and security settings.

Secondary Source Analysis

Theoretical vulnerabilities were discovered to be real failures when testing 52 IoT-management platforms (Ashwag Alotaibi , Huda Aldawghan and Ahmed Aljughaiman, 7 March 2025). Overall, it was found that 13 platforms had a vulnerability with authentication, eight platforms allowed complete account takeover, and 17 platforms were vulnerable to a platform-wide attack. The new attacks, taking advantage of vulnerabilities in the unauthorized access capability, bypassed the ability to disable SIMs and to spoof SIM and device information (Bhaskar Tejaswi, Student Member, IEEE, Mohammad Mannan, and Amr Youssef, 26 Jul 2023).

Key Challenges to Business Success

User Behavior Gap: The awareness-action gap shows that education is not enough to get users to be secure. In the case of platform businesses, security of the platform needs to be designed with regard to user preferences for convenience, creating a tension between user experience and security enforcement.

Technical Knowledge Deficiency: Lack of technical knowledge was cited as a key challenge to secure device management, according to survey respondents. Installation, authentication by multiple manufacturers with different user interfaces and confusing installation of the firmware upgrades can make these devices difficult to use. The platforms need to be heavily invested in interface design and supporting infrastructure of the IoT platform.

Infrastructure Challenges: In developing markets, an unreliable power supply and substandard Internet connection makes users unable to install security updates. This environmental barrier goes beyond the capabilities of traditional cybersecurity solutions, and will require platforms to be designed to be resilient with infrastructure changes and not assume a stable deployment environment.

Economic Limitations: Expensive secure devices would not be economically feasible for users to upgrade to for better authentication features. The low cost of the devices in emerging markets, and the fact that they are very outdated, requires platforms to address the limited computational power of legacy systems, thus making it impossible to implement advanced authentication methods.

Ecosystem Fragmentation: Each device manufacturer has its own system of authentication, password management policies and firmware update procedure. This disintegration results in confusion to the users, implementation errors and revelation of other fronts of weakness (Bhaskar Tejaswi, Student Member, IEEE, Mohammad Mannan, and Amr Youssef, 26 Jul 2023).

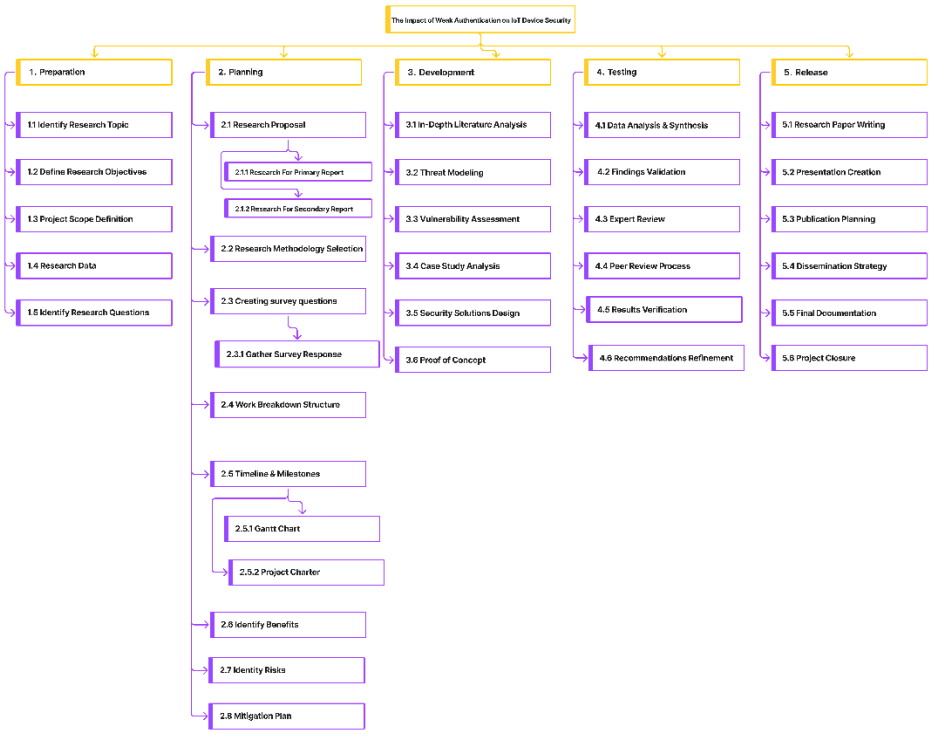
Regulatory Gaps: There's not enough legal protection for digital assets in developing regions, reducing the likelihood of implementing expensive protective measures. Internet retailers are confronted with varying laws and regulations, costs and complexity on international platforms.

Project Planning: Work Breakdown Structure and Gantt Chart

Work Breakdown Structure

There are 5 phases in this work breakdown structure. They are preparation, planning, development, testing and release.The Preparation phase sets the framework for the project by setting the research topic, research goals, project scope, research data collection and research questions.

The Planning phase involves preparation of the research proposal, the selection of proper research methodology, design of the survey instrument, survey distribution and collection, and setting up the Work Breakdown Structure and timeline of the project along with the Gantt chart and project charter. The Development phase is about performing detailed literature analysis, threat modeling in the context of cybersecurity, vulnerability assessment of IoT systems, analysis of relevant case studies, design of security solutions, and Proof of Concept (PoC) generation. The Testing phase validates the research by analyzing and synthesizing the data, validating the findings, using an expert review process, a peer review process, verifying the results, and using refinement of recommendations. The Release phase controls the completion of the project by completing research paper writing, developing a presentation, planning for publication, establishing a dissemination strategy, final documentation and formal project closure.



Gantt Chart

Preparation and Planning phases take place within the first weeks, with overlapping activities, therefore being able to conduct research proposal development in parallel to the development of the survey questions.

All project elements are complete in the Planning phase before the project moves to 'Development' activities, which finishes on 23/06/2026. Current status of the projects shows that the first two phases have been executed and the following phase (Development, Testing, and Release) will be conducted after the current reporting period. The Gantt chart displays realistic task durations ranging from 10 to 45 days, and includes dependencies that ensure that tasks that need to be done first are done before tasks that depend on them are started.

WBS NUMBER	TASK TITLE	TASK OWNER	START DATE	DUE DATE	DURATION	PCT OF TASK COMPLETE
1	PREPARATION					
1.1	Identify Research Topic	Project Manager	5/8/26	6/23/26	45	100%
1.2	Define Research Objectives	Project Assistant	5/8/26	6/23/26	45	100%
1.3	Project Scope Definition	Project Assistant	5/8/26	6/23/26	45	100%
1.4	Research Data	Project Assistant	5/8/26	6/23/26	45	100%
1.5	Identify Research Questions	Project Manager	5/8/26	6/23/26	45	100%
2	PLANNING					
2.1	Research Proposal	Project Assistant	5/8/26	6/23/26	45	100%
2.1.1	Research For Primary Report	Project Assistant	5/8/26	6/23/26	45	100%
2.1.2	Research For Secondary Report	Project Assistant	5/8/26	6/10/26	32	100%
2.2	Research Methodology Selection	Project Assistant	5/8/26	6/23/26	45	100%
2.3	Create Survey Questions	Project Assistant	5/22/26	6/23/26	31	100%
2.3.1	Gather Survey Response	Project Assistant	5/27/26	6/23/26	26	100%
2.4	Work Breakdown Structure	Project Assistant	6/11/26	6/23/26	12	100%
2.5	Timeline & Milestones	Project Assistant	6/12/26	6/23/26	11	100%
2.5.1	Gantt Chart	Project Assistant	6/12/26	6/23/26	11	100%
2.5.2	Project Charter	Project Assistant	6/12/26	6/23/26	11	100%
2.6	Identify Benefits	Project Assistant	6/13/26	6/23/26	10	100%
2.7	Identify Risks	Project Assistant	6/13/26	6/23/26	10	100%
2.8	Mitigation Plan	Project Assistant	6/13/26	6/23/26	10	100%
3	DEVELOPMENT					
3.1	In Depth Literature Analysis	Project Assistant			0	0%
3.2	Threat Modeling	Cybersecurity/IoT Faculty Advisor			0	0%
3.3	Vulnerability Assessment	IoT Security Specialist			0	0%
3.4	Case Study Analysis	Data Analyst			0	0%
3.5	Security Solutions Design	Project Manager			0	0%
3.6	Proof of Concept	IoT Security Specialist			0	0%
4	TESTING					
4.1	Data Analysis and Synthesis	Data Analyst			0	0%
4.2	Finding Validation	Project Manager			0	0%
4.3	Expert Review	Cybersecurity Consultant			0	0%
4.4	Peer Review Process	Academic Peer Reviewers			0	0%
4.5	Results Verification	Cybersecurity/IoT Faculty Advisor			0	0%
4.6	Recommendation Refinement	Project Manager			0	0%
5	RELEASE					
5.1	Research Paper Writing	Project Manager			0	0%
5.2	Presentation Creation	Project Assistant			0	0%
5.3	Publication Planning	Project Manager			0	0%
5.4	Dissemination Strategy	Department Coordina			0	0%
5.5	Final Documentation	Project Assistant			0	0%
5.6	Project Closure	Project Manager			0	0%

Risks and Benefits of the Proposed Project

Project Benefits

The following proposed project is of great value to the organization of the Data Science Department, the University and the cybersecurity position of the University. This research not only aids in understanding the challenges of IoT authentication but also offers the university a comprehensive insight into the vulnerabilities of this type of authentication and their potential impact on the security of digital security systems, contributing directly to the university's goal of securing digital security systems. The project reveals the following limitations of all the IoT implementations discussed in the project so that the institution could focus the activities that are to be performed to overcome the limitations, and allocate its budget to improve its security.

If successful then the impact will be based on rationale recommendations on better infrastructure, policies and procedures to mitigate risk to academic and financial systems as students' data is vulnerable to expensive breaches, in addition to improving infrastructure and policies. The research helps academics understand the methods of authentication that are used in resource-limited settings for the Internet of Things (IoT), and offers the university to add to the knowledge it helps to provide to the cyber security community. Further, the project is grounded in the development of a research replicable model which can be used for further security assessment in the future to gain institutional capacity for ongoing vulnerability assessment. These findings will alleviate stakeholder worries over the university's data security efforts and may inform industry practices on the topic as they determine how to implement it in a higher education context.

Project Risks

Hands-on Security Testing of an actual IoT platform can be challenging with limited technical resources and the outcomes of the survey and secondary research may not be as detailed or in-depth and usable when applied. There's a risk for the

poor return rates on the survey (46 responses), as primary data may not be of high quality and findings not be generalizable; actual return rate on survey is 46%.

These low rates make for a significant risk: of low quality in the primary data, as only 46 of the surveys were returned, and may not actually be generalizable; survey return rate is 46.

Cybersecurity threats are constantly changing, and research findings may change even before they are complete, new attacks emerge that tackle old ones and/or plug newly discovered vulnerabilities. If more than one stakeholder is involved (e.g. faculty advisor(s) and administrator(s) or external security consultant(s) etc.) there is the risk that when stakeholders have conflicting priorities there is a risk of dependency between them. Due to access limitations for sensitive university systems, the practical testing of the vulnerability is not possible and therefore severely constrains practical testing of the recommendations of the research. Currently, the scope creep is huge risk as IoT, authentication and cyber security intersect in a broad range of ways and without direction on the research it can be very difficult to accomplish this project on time. Finally, institutions and/or budget limitations may prevent recommendations from being adopted, limiting the impact of the research on the day-to-day functioning of the university.

Project Recommendations Against Organizational Needs

The project's suggestions related to choosing the authentication mechanism, Infrastructural limitations and problems indicate the importance of these suggestions in the operational context of the university.

The findings of the evaluation show that research results are relevant on various points to organizational needs. The main analysis of the data to identify gaps between awareness and action is directly relevant to the needs of institutional training and development of institutional policies, the department can design realistic security interventions, taking into consideration real user capabilities and constraints. The department's rationale behind investing in infrastructure and

system improvements is backed up by clear evidence point of failures to authenticate as identified through scientific secondary research of the vulnerabilities of the platforms. The results of the research indicate that the findings from the construction barriers to implementing security found in the research, such as instability of electricity, unreliability of the Internet, pest attacks, etc., are impacted by the universities operational environment and barriers. Technical knowledge gaps recommended are harmonized with institutional needs for enhancement of the user support systems and awareness of security.

But there are restrictions on applicability of recommendations. The research concentrates on aspects of IoT device authentication, while the university's needs, as a specific user base and the specific infrastructure it employs, call for a focus on guidance to use with these specific components. Based primarily on information taken from surveys and secondary literature, their recommendations may lack the technical level and/or actionability necessary to implement without further institutional assessment and adaptability. Where the project focuses on articulating challenges and potential obstacles it might face, the nature of the solutions it can produce can be more detailed than the department needs to get from recommendations to solutions. Additionally, emphasis is lacking on cost benefit analysis to better present the financial support for recommended investments of security improvements. Organizational alignment must be undertaken separately as there is no university-wide consideration of recommendations being made against current systems, policies and governance.

Overall, project recommendations sufficiently address the central components of the needs identified by the organization; namely, understanding vulnerabilities concerning authentication, documenting challenges to implementation in a secure way, and offering analysis specific to target sectors and are applicable to settings. The recommendations will assist with the security priorities and the allocation of resources in intuitions. Recommendations, however, would require further institutional and technical study and modification for research results to be turned

into security improvements at the operational level to address the needs of the university's institutional needs and technical infrastructure.

Evaluation of Project Planning Recommendations and Research Reliability

A planned stage for literature analysis, threat modeling, threat assessment and design of security solutions systematically addresses the need to handle the literature analysis, evaluation of authentication vulnerabilities and security implications of these vulnerabilities in detail and systematically: A planned step for literature analysis, threat modeling, threat assessment and design of security solutions systematically addresses the need to handle literature analysis, evaluation of authentication vulnerabilities and security implications of these vulnerabilities in detail and systematically. The twelve weeks sets realistic timelines for research and recommendation to be presented to the department, thus helping fulfill the department's aim to improve digital security systems within a predetermined time frame of a project. There are capacity restrictions for who they can assign to the project manager, data analysts, faculty advisors and security specialists, and there is a quality assurance procedure in place that is overseen.

However, there are disparities between the needs of project plans and the implementation in the organization. The project plan identifies the Research and Analysis phases and fewer resources to Testing and Release phases, where the research outcome is applied to provide recommendations for institutional deployment. Time spent on Preliminary research is dominated in Week 6 giving the role of low levels of time compared to the other roles, whereas Development and Testing has not such high amount of time allocated to it but relatively less than the other roles in complexity, especially recommendations. Although the project charter and the cost analysis are listed as deliverables for the project, few details are included in the Gantt chart, and there appears to be limited planning on budgeting and justification for financial reasons to support the actions taken by the organization. Likewise multiple stakeholders, such as IT administrators, leadership

and faculty, are not explicitly communicated in the project plan about the dissemination of the project's research outcomes, increasing the risk that recommendations are only in research documentation, and not part of the process of creating operational changes.

Evaluation of Research Accuracy and Research Reliability

This research method is a mix of reliability, accuracy and must be interpreted carefully. The top graphic segment has secured 46 respondents from Myanmar who are of various age groups including student, employed professional and self-employed and got qualitative data on the practice of user authentication and users' awareness of security. The number of respondents of this survey, however, is limited and only a small number can make comparisons between the various demographic segments, or make generalizations on a larger population of users. The survey tool assesses subjects' self-reported behavior, rather than how they actually behave, which may lead to some bias as the respondents may feel that they are doing things that they believe would reflect a security way of doing things, but they are not actually doing. The hypothetical and/or future question items in surveys are less reliable, and can produce an exaggerated level of security awareness/risk mitigation among respondents.

The secondary research part on the vulnerabilities of authentication shows increased reliability due to the use of peer-reviewed scientific papers and empirical security tests of 52 IoT management platforms. All the academic sources are analyzing the roles of authentication mechanisms and attack vectors and published in reputable journal and well documented systematic literature review. The empirical security evaluation included real-world platforms with real-world conditions, and confirmed the broken authentication and unauthorized access and remote code execution exploits. These secondary sources, however, tend to be more general on the vulnerabilities of an IoT Platform as a whole and less particular to the university context, and so the findings have to be extrapolated to the university context.

In research accuracy, there are certain limitations that result from the limitations of the methods used. Although the project combines the concept of IoT device authentication with IoT platform management authentication, it sometimes conflates the different types of device authentication and management authentication required by connected devices with authentication and management of a fleet of devices. The conceptual ambiguity has impact on the accuracy of recommendation in institutional use. The research doesn't evaluate the vulnerabilities of the existing infrastructure within the university, but rather gives general sector analysis which can be applied in higher education and enterprise environments. These are initial research data collected at opportune times and at specific locations (Myanmar) and the applicability of infrastructure barriers and economic factors affecting the operational environment of the university are a matter of question, as the conditions in the developing markets listed in the surveys may be very different.

Overall Evaluation

Project planning is structured and will be implemented to develop a structured research process with distribution of resources during the project phases, reflecting a good project management approach. Primary survey findings and secondary academic and empirical evidence used in the research provides additionally triangulation among various types of evidence. The research reliability is, however, limited due to the modest primary sample size, the recall of respondents' behaviors and the time-specificity of administration of the survey.

Secondary research has good academic rigor, and is a research project that requires interpretation to apply to institutions. The project recommendations would benefit from expressing research limitations with confidence intervals for the empirical claims made, and having further organizational evaluations before proposing infrastructure changes. In general, the project planning and research serve as useful background information that will inform the University's cyber

security enhancement goals; however, they need to be validated and tailored to suit the specific needs of the University's technical infrastructure and organizational needs.

Conclusion

In conclusion, it could be said that this study revealed that 93.5% of the users know about the password policies but 43.5% of them do not bother changing their password because the region's electric and network power is unstable so that the process of changing the password is hindered. The project plan is 12 weeks long, but does not provide sufficient time for testing, deployment and a financial budgeting, and it does not start the research process early enough. The first results obtained from the small number of 46 need to be adapted to the infrastructure and financial capacities of the University before they can be used – this is what is meant by future developments.

References

Ashwag Alotaibi , Huda Aldawghan and Ahmed Aljughaiman, 7 March 2025. A Review of the Authentication. *Sensors,* p. 43.

Bhaskar Tejaswi, Student Member, IEEE, Mohammad Mannan, and Amr Youssef, 26 Jul 2023. Security Weaknesses in IoT Management Platforms. *IEEE Internet of Things Journal,* VOL. 00(No. 0), p. 16.